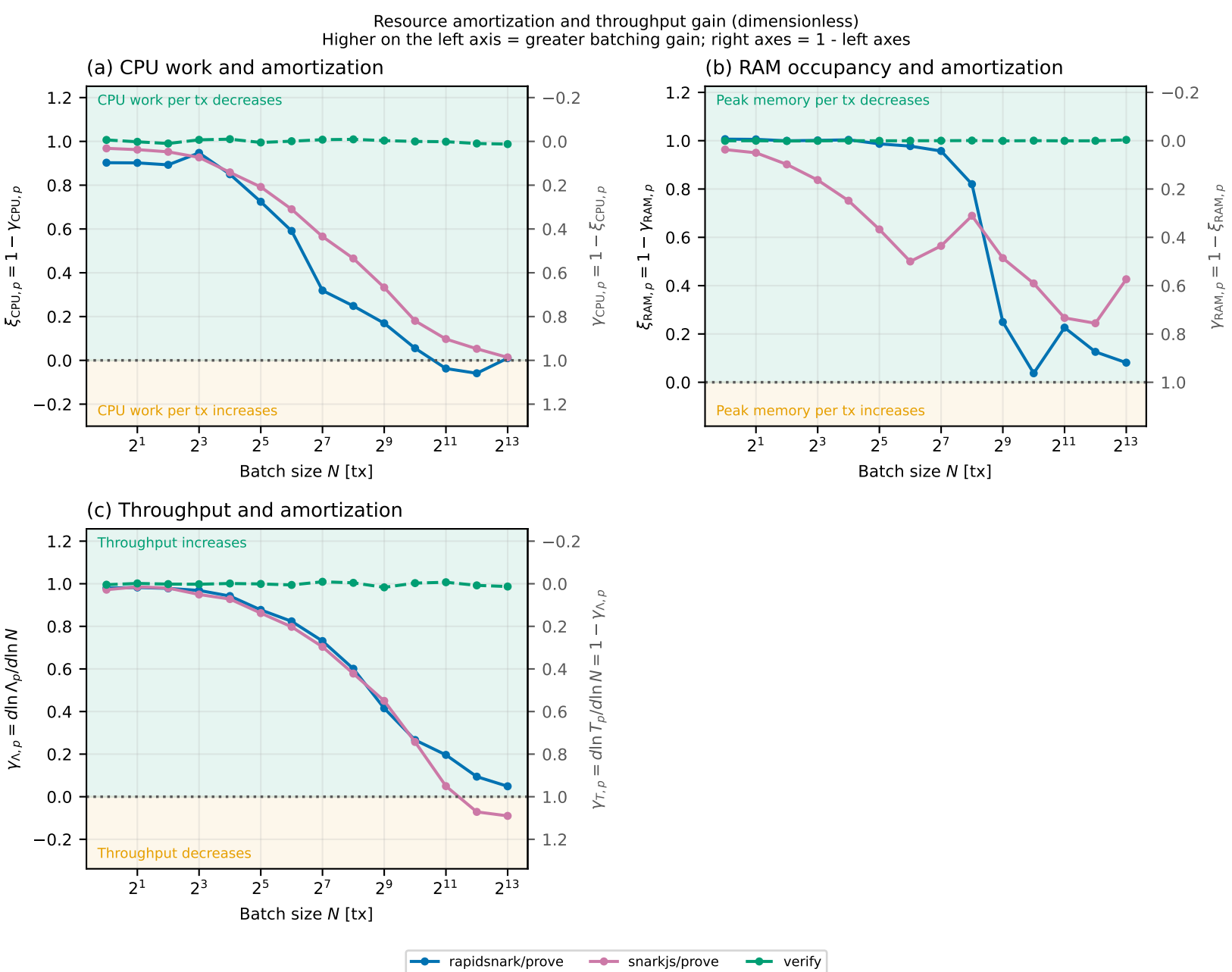
Resource amortization and throughput gain (dimensionless)
Higher on the left axis = greater batching gain; right axes = 1 - left axes
(a) CPU work and amortization
CPU work per tx decreases
CPU work per tx increases
$\xi_{\mathrm{CPU},p} = 1 - \gamma_{\mathrm{CPU},p}$
$\gamma_{\mathrm{CPU},p} = 1 - \xi_{\mathrm{CPU},p}$
Batch size N [tx]
(b) RAM occupancy and amortization
Peak memory per tx decreases
Peak memory per tx increases
$\xi_{\mathrm{RAM},p} = 1 - \gamma_{\mathrm{RAM},p}$
$\gamma_{\mathrm{RAM},p} = 1 - \xi_{\mathrm{RAM},p}$
Batch size N [tx]
(c) Throughput and amortization
Throughput increases
Throughput decreases
$\gamma_{\Lambda,p} = d\ln\Lambda_p/d\ln N$
$\gamma_{T,p} = d\ln T_p/d\ln N = 1 - \gamma_{\Lambda,p}$
Batch size N [tx]
rapidsnark/prove
snarkjs/prove
verify